From complete channel order to capacity and one-way distillation*

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Abstract

A flagged channel with a four-dimensional auxiliary input turns every failed complete information comparison into positive private and quantum capacity. This transfers complete-order undecidability to exact-zero capacity and one-way distillation, even under any prescribed two-way distillation lower bound. All reference transmission and activation use the constructed channel itself.

1 Introduction

Complete information comparison permits an experimenter to keep a quantum reference correlated with a channel input. Communication does not give the receiver such a reference for free. An information advantage visible on a large joint system may disappear when the receiver sees only the channel output. Turning a failed comparison into a positive communication rate therefore requires a way to send the witness itself.

Our construction does this with a fixed four-dimensional auxiliary input. A fair branch flag routes that input either to the receiver or to the environment, while selecting complementary outputs of the source channels. Further uses distribute an arbitrarily large, but finite, reference in quantum shares. The difficulty is not just recovering it: unsuccessful routing patterns could overwhelm a small information advantage. We distribute a classical key as well, arranging that it usually becomes available only after enough quantum shares have arrived. All other patterns in which the key is hidden contribute exactly zero. This produces a finite block with positive private information.

The same channel contains a qubit half erasure. Although half erasure alone has zero quantum capacity, the Smith–Yard identity converts a finite private advantage into positive coherent information when an appropriate finite helper is supplied. We realize that helper from further copies of the constructed channel. Thus the entire argument uses ordinary unassisted communication through one memoryless channel. Passing to its Choi state then transfers positivity to one-way entanglement and key distillation.

Every step preserves *strict positivity*, with its actual resource cost charged. The witnessing reference, blocklength and positive rate depend on the source channels. The resulting undecidability is exact zero versus positive capacity; it does not supply a uniform positive rate gap or an obstruction to additive capacity approximation.

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1.1 The channel and the main theorem

All spaces are finite dimensional. A channel is a completely positive trace-preserving linear map. For a Stinespring isometry $V : A \to B \otimes E$, its receiver and complementary channels are

$$L(\rho) = \mathrm{Tr}_E\, V\rho V^*, \qquad L^c(\rho) = \mathrm{Tr}_B\, V\rho V^*.$$

We always use genuine complements of this form; enlarging an output by an isometry does not change any comparison or capacity below. A pure-state symbol without a ket also denotes its rank-one projector.

Let M, N have common input A. We write $N \succeq^{\mathrm{c}}_D M$ when

$$D\big((N \otimes \mathrm{id}_R)(\alpha)\|(N \otimes \mathrm{id}_R)(\beta)\big) \geq D\big((M \otimes \mathrm{id}_R)(\alpha)\|(M \otimes \mathrm{id}_R)(\beta)\big) \tag{1}$$

for every finite reference R and all states α, β on $A \otimes R$. Here $D(\rho\|\sigma) = \mathrm{Tr}\,\rho(\log_2 \rho - \log_2 \sigma)$, with value $+\infty$ outside support inclusion. The reference is untouched, but α_R and β_R need not be equal. No uniform bound is placed on $\dim R$.

For an integer $z \geq 4$, take an auxiliary input $Z = \mathbb{C}^z$. The channel chooses a fair branch independently of its input. In one branch Z goes to Eve, and in the other it goes to Bob, as shown in Figure 1. Both parties receive the branch flag; the sender cannot choose it.

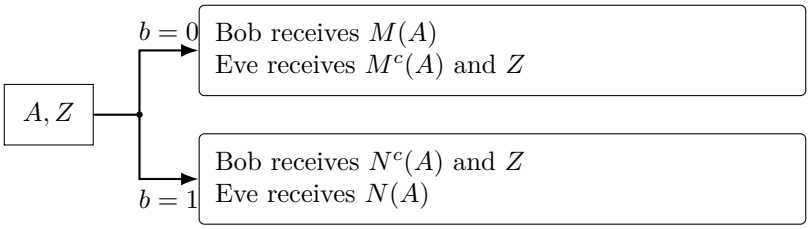


Figure 1: One use of the compiler. The common flag records which Stinespring branch was used. The auxiliary input reaches exactly one party; with A fixed and its outputs discarded, this is half erasure.

In formulas, the receiver and environmental channels are

$$\mathcal{C}_z(\rho_{AZ}) = \tfrac{1}{2}M(\mathrm{Tr}_Z\,\rho) \oplus \tfrac{1}{2}(N^c \otimes \mathrm{id}_Z)(\rho), \tag{2}$$
$$\mathcal{C}^c_z(\rho_{AZ}) = \tfrac{1}{2}(M^c \otimes \mathrm{id}_Z)(\rho) \oplus \tfrac{1}{2}N(\mathrm{Tr}_Z\,\rho). \tag{3}$$

The orthogonal direct sums carry the flags. Section 2 verifies that these are genuine complementary marginals.

Theorem 1.1 (Universal channel-order compiler)**.** *For every finite common-input pair M, N and every integer $z \geq 4$,*

$$N \succeq^{\mathrm{c}}_D M \quad \Longleftrightarrow \quad P(\mathcal{C}_z) = 0 \quad \Longleftrightarrow \quad Q(\mathcal{C}_z) = 0. \tag{4}$$

If M, N have rational-complex Choi matrices, suitable complements and the Choi matrix of $\mathcal{C}_z$ can be computed exactly over $\mathbb{Q} + i\mathbb{Q}$. Its input dimension is $z \dim A$.

Here Q and P are the ordinary, unassisted quantum and private classical capacities. For an ensemble $\{p_x, \rho_x\}$ and average $\bar\rho = \sum_x p_x\rho_x$, write

$$\chi(X : B) = S(\bar\rho_B) - \sum_x p_x S(\rho^B_x), \qquad I_{\mathrm{c}}(L, \rho) = S(L(\rho)) - S(L^c(\rho)),$$

where $S(\rho) = -\operatorname{Tr}\rho\log_2\rho$. The coding theorems give

$$P(L) = \sup_{n\geq 1}\frac{1}{n}\sup_{\{p_x,\rho_x\}}\left[\chi(X:B^n) - \chi(X:E^n)\right], \tag{5}$$

$$Q(L) = \sup_{n\geq 1}\frac{1}{n}\max_{\rho} I_{\mathrm{c}}(L^{\otimes n},\rho). \tag{6}$$

The ensemble and state in these formulas are inputs to the full block, and may be entangled across its uses. The formulas include mixed ensemble letters. Both capacities are nonnegative. Moreover $Q \leq P$: a spectral pure-state ensemble of ρ has private information $I_{\mathrm{c}}(L,\rho)$. We use these established coding theorems in the form proved by Devetak [2].

The proof has a short zero direction and a constructive positive direction. If $N \succeq_D^{\mathrm{c}} M$, complement reversal makes Eve at least as informative as Bob in the two branch comparisons; tensorization then excludes private information at every blocklength. If the order fails, Section 3 sends a split-key witness and obtains positive private information. Section 4 turns that witness into positive coherent information using only the embedded half erasure.

1.2 Exact-zero classification and distillation

The companion paper [1] proves the complete-order equivalence, complement reversal, and effective order-hardness theorem restated in Section 2. Combining that theorem with our compiler gives the following classification.

Theorem 1.2 (Capacity and one-way distillation)**.** *On valid finite rational-complex channel descriptions, the sets*

$$\{L : Q(L) > 0\}, \qquad \{L : P(L) > 0\}$$

are RE*-complete under computable many-one reductions; their exact-zero sets are* coRE*-complete. On valid finite rational-complex bipartite states, the same classifications hold for positive and zero Alice-to-Bob distillable entanglement $D_{\to}$ and distillable key $K_{\to}$.*

For every prescribed rational $r > 0$, the state hardness reductions can be chosen to satisfy simultaneously

$$\omega_A = I_A/\dim A, \qquad K_{\leftrightarrow}(\omega) \geq D_{\leftrightarrow}(\omega) \geq r. \tag{7}$$

Channel and state dimensions vary with the instance.

The state result follows from a general channel–state connection, proved in Section 5. If $J_L = (\mathrm{id}\otimes L)(\Phi_d)$ is the normalized Choi state of a channel with d-dimensional input, then

$$D_{\to}(J_L) \leq Q(L), \qquad D_{\to}(J_L) > 0 \iff Q(L) > 0, \tag{8}$$

$$K_{\to}(J_L) \leq P(L), \qquad K_{\to}(J_L) > 0 \iff P(L) > 0. \tag{9}$$

These are positivity equivalences, not equalities of rates. The successful filter that prepares a desired channel input from the Choi state may be very unlikely. Our proof accounts for that probability explicitly.

The two-way promise in (7) highlights the role of communication direction. Every compiled Choi state contains an erasure state from which Bob can identify useful entanglement. A backward announcement of the successful positions gives a uniform positive two-way rate. The one-way task does not allow that announcement.

1.3 Relation to prior work

Receiver–environment comparisons have long connected channel orders with private and quantum capacities [6, 7, 8]. The complete relative-entropy/Holevo equivalence and tensorization are part of this established framework [7]. Hirche's recent estimation approach gives a coherent-qubit passage from complete measured χ^2 comparison to complete relative entropy [9]; the reverse implication and complement anti-isomorphism used here are proved in the companion paper [1].

Our private-to-quantum step is the finite Smith–Yard activation identity [10]. Its symmetric-side-channel interpretation appears in [11, 12]. The new issue here is to realize the required helper from further copies of the target channel, with no auxiliary communication resource. We use standard stabilizer erasure correction [13, 14] and binary coset secrecy [15]. Their short finite counting arguments are included because exact hiding and exact recovery are central to the proof. Recent incapacity and activation constructions [19, 20] show why zero capacity alone must not be confused with complete environmental dominance.

The Choi bridges combine ensemble steering, in the sense of Hughston, Jozsa and Wootters [4], with the one-way coding theorems of Devetak and Winter [3, Theorems 8 and 13]. Postselected Choi coherent information is studied explicitly in [21]. The upper bound $D_{\to}(J_L) \le Q(L)$ appears in Bennett et al. [5, Eq. (53)]; the private converse uses the transcript-averaging argument in Devetak's equivalence of private transmission and forward-public key generation [2, Section 3]. We include the elementary steering and averaging proofs because their success probabilities explain why positivity transfers while rates need not agree.

Finally, arbitrarily long superadditivity [23, 24], noncomputability for channels with memory [25], and hardness for succinct quantum channels or restricted assisted zero-error models [26] concern different questions. Here the channels are memoryless, explicitly described by rational matrices, and the capacities use ordinary vanishing error without assistance. The hardness conclusion inherits the complete-order reduction of [1], whose public starting point is $\mathrm{MIP}^* = \mathrm{RE}$ [27].

2 Complete comparison and its capacity consequence

Relative entropy is convenient for stating an order, but communication is expressed in terms of ensembles. The Holevo formulation connects the two and explains why the comparison must be complete: in a many-use code, the other inputs and outputs serve as references during each one-factor replacement.

Lemma 2.1 (Holevo form of complete comparison)**.** *$N \succeq_D^{\mathrm{c}} M$ holds if and only if*

$$\chi(X : NR) \ge \chi(X : MR) \tag{10}$$

for every finite ensemble on AR and every finite reference R. A failure has a strict binary-ensemble witness. Complete comparison tensorizes, including for entangled inputs.

Proof. The identity

$$\chi(X : LR) = \sum_x p_x D(L_R(\rho_x) \| L_R(\bar{\rho})), \qquad L_R = L \otimes \mathrm{id}_R,$$

gives the forward implication. For the converse, fix a faithful denominator β and use the binary ensemble $\{t, \alpha; 1-t, \beta\}$. As $t \downarrow 0$,

$$t^{-1}\chi(\{t, \alpha; 1-t, \beta\} : LR) \longrightarrow D(L_R(\alpha) \| L_R(\beta)).$$

A strict relative-entropy violation therefore gives a strict binary Holevo violation for small enough t. For a nonfaithful β, first replace it by $(1-\epsilon)\beta + \epsilon\pi$ with π faithful. The supported relative-entropy limit preserves every strict violation, including one with infinite larger divergence. This proves the converse and the binary assertion.

For tensorization, replace one channel factor at a time in the Holevo quantity. At each replacement all other inputs, together with outputs already produced, form the untouched reference. Inequality (10) applies at that step; summing the steps proves the product comparison. This is the standard complete less-noisy argument [7, Proposition 2.3 and Lemma 3.1]. □

The following theorem records the precise mathematical input from the companion paper. Its first part is the only nonstandard order property used by the universal compiler; its second part is needed only for undecidability.

Theorem 2.2 (Complete-order duality and hardness [1])**.** *For finite channels M, N with the same input and genuine complements,*

$$N \succeq_D^{\mathrm{c}} M \quad \Longleftrightarrow \quad M^c \succeq_D^{\mathrm{c}} N^c. \tag{11}$$

There is a total computable map from a Turing machine T, with its input incorporated, to rational-complex common-input channels M_T, N_T such that

$$T \text{ does not halt} \quad \Longleftrightarrow \quad N_T \succeq_D^{\mathrm{c}} M_T. \tag{12}$$

Consequently complete comparison of rational channels is coRE*-complete.*

The reference convention is exactly that of (1): the two marginals may differ. Complement reversal follows by identifying complete relative-entropy and SLD orders and applying a common purification identity. Appendix A includes the supported purification proof. The full equivalence of metrics and the nonlocal-game reduction (12) are proved in the companion.

The routing has been chosen so that Bob's excess information splits into two complete-order differences. One compares M with N after discarding Z; the other compares N^c with M^c while retaining Z. Complement reversal gives both differences the same sign.

Proposition 2.3 (The zero direction)**.** *Equations* (2)–(3) *are complementary marginals of one physical channel. If $N \succeq_D^{\mathrm{c}} M$, then $\mathcal{C}_z^c \succeq_D^{\mathrm{c}} \mathcal{C}_z$ and $P(\mathcal{C}_z) = Q(\mathcal{C}_z) = 0$.*

Proof. In branch zero apply a Stinespring isometry of M and route Z to Eve. In branch one apply one of N, interchange its receiver and environment outputs, and route Z to Bob. Place the two branches in orthogonal subspaces at both outputs. The balanced sum of the two branch isometries is an isometry; both partial traces remove the cross terms and give the displayed channels.

For any ensemble on AZR, the branch probabilities are independent of its label. Holevo information therefore averages over the branches:

$$\begin{aligned}&\chi(X:\mathcal{C}_z R) - \chi(X:\mathcal{C}_z^c R)\\ &\quad = \tfrac{1}{2}\big[\chi(X:MR) - \chi(X:NR)\big] + \tfrac{1}{2}\big[\chi(X:N^cZR) - \chi(X:M^cZR)\big] \le 0.\end{aligned}$$

The first comparison uses the marginal after tracing out Z; the second retains Z as part of the reference and uses (11). Lemma 2.1 gives complete environmental dominance. Its tensorization excludes positive private information at every blocklength, so (5) gives $P = 0$, and $Q \le P$ gives $Q = 0$. □

3 Sending the reference and unlocking the witness

Suppose $N \not\succeq^{\mathrm{c}}_{D} M$. Complement reversal and Lemma 2.1 give a finite binary ensemble

$$\{p_x, \rho_x^{AR}\}_{x\in\mathbb{F}_2}, \qquad \delta = \chi(X : N^c R) - \chi(X : M^c R) > 0. \tag{13}$$

Since the label is binary, $0 < \delta \le 1$. The advantage is available on the joint output and R, so the reference is part of the resource that must reach the receiver. Fix the witness and embed R into $k = \lceil \log_2 \dim R \rceil$ qubits, allowing $k = 0$.

Take $z = 4$ and write $Z = q \otimes c$, two qubits. One active slot carries A and one classical key share. Further slots carry quantum shares of R through q and classical shares of a second key bit through c. A slot routes its two shares together. We shall place the classical revelation threshold slightly above the quantum recovery threshold, so that a party lacking a usable reference usually lacks the key as well. The logical size k is fixed throughout this coding argument; only the number of helper slots grows.

3.1 A split key removes the unwanted terms

Let U be the uniform message bit. The elementary masked ensemble

$$\sigma_u^{ARK} = \sum_x p_x \rho_x^{AR} \otimes |x+u\rangle\langle x+u|_K$$

has AR marginal independent of u. With the key present, a block entropy calculation gives, for every channel L on A,

$$\chi(U : LRK) = 1 - H(p) + \chi(X : LR). \tag{14}$$

Here $H(p) = -\sum_x p_x \log_2 p_x$. The average key is uniform and independent of the average LR state, while each letter has entropy $H(p) + \sum_x p_x S(L_R(\rho_x))$. The bias term $1 - H(p)$ is harmless because it cancels between the two channels.

A single key would be insufficient: it would leave the party on the wrong active branch in the information balance. We want one share tied to the active branch and another tied to recovery of the reference. Neither share alone should reveal anything, even jointly with AR. Split the mask into two shares. Choose an independent uniform bit v and set $K_0 = v$, $K_1 = x + u + v$, with addition in $\mathbb{F}_2$:

$$\tau_u = \sum_{x,v} \frac{p_x}{2} \rho_x^{AR} \otimes |v\rangle\langle v|_{K_0} \otimes |x+u+v\rangle\langle x+u+v|_{K_1}. \tag{15}$$

Either key share alone hides the message even from someone holding all of AR:

$$\tau_u^{ARK_j} = \bar{\rho}^{AR} \otimes I_{K_j}/2 \qquad (j = 0, 1). \tag{16}$$

When both shares are present, relabeling them as $(K_0, K_0 + K_1)$ leaves the original mask and independent uniform junk. Thus

$$\chi(U : LRK_0K_1) = 1 - H(p) + \chi(X : LR). \tag{17}$$

The first share will select the party that can see the message on the active branch. The second will select the helper masks on which we can use the reference.

A Bell-state example. Take $N^c = \mathrm{id}$ and let $M^c = \Delta$ dephase a qubit in the computational basis. Use the equiprobable letters

$$|\phi_x\rangle_{AR} = \frac{|00\rangle + (-1)^x|11\rangle}{\sqrt{2}}, \qquad x \in \mathbb{F}_2.$$

Their A marginals are identical, so A alone carries no label information. With R present the identity output distinguishes the two Bell states perfectly, whereas

$$(\Delta \otimes \mathrm{id}_R)(\phi_x) = \tfrac{1}{2}|00\rangle\langle 00| + \tfrac{1}{2}|11\rangle\langle 11|$$

is independent of x. Hence $\delta = 1$. Given both shares $K_0 = v$, $K_1 = x + u + v$, the identity receiver measures the Bell label x and reads $u = x + K_0 + K_1$. With either share missing, the remaining uniform pad hides u. With both shares but no reference, the identical A marginals still hide it. Even with the reference and both shares, the dephased output has no message information.

Here one helper slot suffices: send R and K_1 directly through its Z input, fixing its A input pure. There are four equiprobable active and helper branch patterns. In pattern $(1, 1)$ Bob obtains one message bit and Eve obtains none. In $(0, 0)$ Eve receives both shares and R, but dephasing has erased the Bell label. The other two patterns split the key between the parties, so both learn nothing. The two-slot block therefore has private information $1/4$ bit, giving $P(\mathcal{C}_4) \geq 1/8$ bit per use for this channel pair.

Return to the arbitrary witness (13). Use one active slot and n helper slots as follows. The active A contains the A part of (15), its c contains K_0, and its q is fixed pure. Isometrically encode R into n qubit shares and classically encode K_1 into n binary shares. Helper i contains these shares in q_i, c_i; its A_i is fixed pure. All preparation is sender-side, before any route flags are generated.

Let $S \subseteq [n]$ be the helper slots whose Z reaches Bob. Every mask has probability 2^{-n}; Eve receives the complementary shares. Write Q_S, C_S for the raw quantum and classical shares. Conditioned on the route flags, the fixed-input helper payloads are independent ancillary states and do not affect the information.

On active branch one, Bob receives $N^c(A)K_0$, whereas Eve lacks K_0 and has zero U-information by (16). On branch zero, Eve receives $M^c(A)K_0$, whereas Bob's state is independent of U. Pairing the second branch's mask with its complement gives the exact ordinary private information of the block:

$$\mathrm{PI}_n = \frac{1}{2}\,\mathbb{E}_S\, b(S), \qquad b(S) = \chi(U : N^c(A)Q_S K_0 C_S) - \chi(U : M^c(A)Q_S K_0 C_S). \tag{18}$$

The equality uses the symmetry of the physical mask distribution: S and its complement have the same probability. Thus the two comparisons contain the same *raw* survivor channel. No decoding or postselection has entered this private-information identity.

There are now two exact secrecy mechanisms. A party missing K_0 learns nothing, regardless of its quantum shares. If C_S hides K_1 jointly with all correlated systems, even the party holding K_0 learns nothing. The only dangerous masks reveal K_1 without permitting recovery of R. We cannot exclude every such mask; instead we make their total probability small enough that their possible loss is outweighed by the good masks' advantage.

3.2 Two elementary sharing codes

The classical code is a random binary coset code. Choose a matrix $G \in \mathbb{F}_2^{n\times r}$ and vector $g \in \mathbb{F}_2^n$, and let T be r independent uniform random bits. Encode

$$C = GT + gK_1. \tag{19}$$

For a mask S, the conditional distributions are uniform on the two cosets $\operatorname{im} G_S$ and $g_S + \operatorname{im} G_S$. Consequently they are either identical or disjoint: that mask hides the key exactly or reveals it exactly. This all-or-nothing statement is useful here in a way that small average leakage alone would not be.

Lemma 3.1 (Binary shares)**.** *If $g_S \in \operatorname{im} G_S$, then C_S is independent of K_1. If $g_S \notin \operatorname{im} G_S$, an invertible classical relabeling makes C_S equal to K_1 together with independent uniform junk. For independently uniform G, g and $s = |S|$,*

$$\Pr_{\text{codes}}[\textit{key revealed} \mid |S| = s] \le 2^{s-r} \qquad (s \le r), \tag{20}$$

$$\Pr_{\text{codes}}[\textit{key hidden} \mid |S| = s] \le 2^{r+1-s} \qquad (s \ge r+1). \tag{21}$$

Proof. In the disjoint-coset case choose a linear functional vanishing on $\operatorname{im} G_S$ and taking value one on g_S, then complete it to linear coordinates. The first coordinate is K_1 and the others are uniform independently of it. In the identical-coset case the share channel is a replacer, so the independence remains true jointly with any quantum system correlated with K_1.

Full row rank of G_S implies hiding. When $s \le r$, each nonzero left-kernel candidate has probability 2^{-r} of annihilating G_S; a union bound over fewer than 2^s candidates gives (20). Full column rank of $[G_S\ g_S]$ implies revelation. A union bound over its fewer than 2^{r+1} nonzero right-kernel candidates gives (21). This is the basic coset-secrecy mechanism of linear wiretap coding [15]. □

The quantum code is a random stabilizer code of fixed logical size. Recall that n-qubit Pauli operators modulo phase form the symplectic space $\mathbb{F}_2^{2n}$. An $[[n, k]]$ stabilizer code corresponds to an isotropic subspace L of dimension $n - k$; its Pauli normalizer is $L^\perp$, of dimension $n + k$. Apply a uniformly random Clifford to one such code. For a lost set $T \subseteq [n]$, call the mask *good* if no nonzero Pauli vector supported on T lies in $L^\perp$. This is a sufficient certificate, not a characterization of every correctable erasure pattern.

Lemma 3.2 (Exactly reversible quantum shares)**.** *Every good mask permits exact recovery of the full logical input from the surviving raw shares. For a fixed lost set of size e, a random code satisfies*

$$\Pr[\textit{not good}] \le (4^e - 1)\frac{2^{n+k} - 1}{2^{2n} - 1} \le 2^{2e-n+k+1}. \tag{22}$$

On a good mask, replacing R by the surviving raw shares preserves every joint Holevo quantity involving R and arbitrary correlated systems.

Proof. Clifford transitivity implies that every nonzero Pauli vector belongs to the random $L^\perp$ with probability $(2^{n+k} - 1)/(2^{2n} - 1)$. There are $4^e - 1$ nonzero vectors supported on the lost set. Their union bound proves the first inequality; bounding numerator and denominator proves the second.

Every nonidentity lost-system Pauli on a good set anticommutes with some stabilizer, so its compression to the code is zero. The lost channel is therefore exactly $\rho \mapsto \pi_T \operatorname{Tr} \rho$. To see recovery directly, expand the encoding isometry as

$$W|i\rangle = 2^{-e/2} \sum_t |v_{i,t}\rangle_S |t\rangle_T.$$

The lost-channel identity implies $\langle v_{j,t'}|v_{i,t}\rangle = \delta_{ij}\delta_{tt'}$. Hence the survivor channel has the form

$$\rho \longmapsto V(\rho \otimes \pi_T)V^* \tag{23}$$

for an isometry V, and has a trace-preserving inverse on the logical input. This is the exact erasure-correction condition [14, 13]. Applying data processing first to the survivor channel and then to its recovery proves equality of the joint Holevo quantities, including when the logical input is correlated with A or a classical label. □

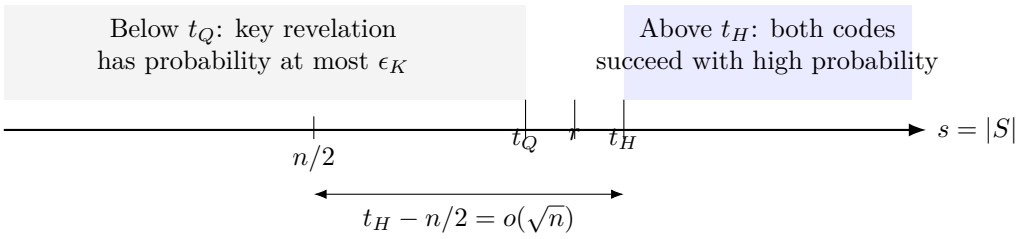


Figure 2: The key threshold is placed above the quantum recovery threshold, but all thresholds lie within $o(\sqrt{n})$ of the binomial center. The probability bounds average over codes; they are not deterministic threshold guarantees for a fixed code. For every individual code and mask, the key is exactly hidden or exactly revealed.

3.3 Synchronizing revelation and recovery

Exact hiding and exact recovery are what make the average manageable. We never need to estimate a small information leak on every bad mask. The bracket in (18) has just three cases:

Mask condition	$b(S)$
Classical key hidden	0
Classical key revealed, quantum mask good	δ
Classical key revealed, quantum mask not good	at least -1

The first row follows from exact hiding and (16). The second follows from reversible raw shares and (17). The last uses $0 \le \chi(U:\cdot) \le H(U) = 1$.

The thresholds must satisfy two competing requirements. Revealing the key should be rare below the quantum recovery window, yet the joint success region must retain positive physical probability. A separation growing like $n^{1/4}$ meets both: the coding errors decay exponentially in that separation, while the entire window is negligible on the binomial fluctuation scale $\sqrt{n}$.

Let $a_n = \lceil n^{1/4} \rceil$ and set

$$\begin{aligned} t_Q &= \lceil (n+k)/2 \rceil + a_n, & r &= t_Q + a_n, & t_H &= r + 1 + a_n, \\ \epsilon_Q &= 2^{1-2a_n}, & \epsilon_K &= 2^{-a_n}, & p_H &= \Pr[\mathrm{Bin}(n,1/2) \ge t_H]. \end{aligned} \tag{24}$$

Take n large enough that all thresholds are at most n. For a fixed survivor size below t_Q, the probability over the classical code of revealing the key is at most ϵ_K. At size at least t_Q, the quantum-failure probability is at most ϵ_Q. At size at least t_H, the key-hiding probability is at most ϵ_K. These assertions follow directly from Lemmas 3.1 and 3.2.

Averaging over the independent codes and physical mask, a revealing good mask has probability at least $p_H - \epsilon_K - \epsilon_Q$. A revealing nongood mask has probability at most $\epsilon_K + \epsilon_Q$: below t_Q revelation is rare, and above it quantum failure is rare. The three-case table therefore gives

$$\mathbb{E}_{\text{codes}}\, \mathrm{PI}_n \ge \frac{1}{2}\left[\delta(p_H - \epsilon_K - \epsilon_Q) - \epsilon_K - \epsilon_Q\right]. \tag{25}$$

Both errors vanish. Since k is fixed, $t_H - n/2 = k/2 + O(n^{1/4}) = o(\sqrt{n})$, and $p_H \to 1/2$. One elementary way to see the latter is to use symmetry and

$$\max_j \Pr[\mathrm{Bin}(n,1/2) = j] \le (\lfloor n/2 \rfloor + 1)^{-1/2}.$$

For $n = 2m$ the bound follows by induction from the ratio $(2m+1)/(2m+2)$ of successive central binomial probabilities; the odd case has no larger maximum atom. Moving the central threshold by $o(\sqrt{n})$ therefore changes the upper-half probability by $o(1)$.

The lower bound in (25) tends to $\delta/4 > 0$. Some finite n and one fixed pair of encoders have $\mathrm{PI}_n > 0$, whence

$$P(\mathcal{C}_4) \geq \frac{\mathrm{PI}_n}{n+1} > 0. \tag{26}$$

The expectation selects an encoder before transmission. It does not condition the rate on favorable route masks: every mask appears in the actual channel block. The exact zeros below the revealing window are what allow the small positive witness to survive averaging.

For $z > 4$, embed $q \otimes c$ in any fixed four-dimensional subspace of Z. The receiver channels on this subspace are isometric copies of those just used. Thus $P(\mathcal{C}_z) > 0$ as well. This completes the private-capacity part of Theorem 1.1.

4 An embedded half erasure converts privacy into coherence

Private information is a receiver's advantage about a classical label. Positive coherent information asks for more: the quantum purification of the transmitted state must remain useful. The Smith–Yard construction supplies the missing purification through half erasure. In the two erasure branches, the label-entropy terms cancel, leaving half the original private advantage.

The helper may have a large dimension. A single auxiliary qubit cannot carry it, but the helper is fixed once the private witness is fixed. Many qubit half erasures realize an exact erasure of that fixed system with transmission probability approaching $1/2$. The logical rate of this realization tends to zero; a finite realization is nevertheless enough to preserve a strict coherent-information advantage.

Throughout this paper the parameter of an erasure channel is its *transmission* probability:

$$\mathcal{E}_{d,p}(\rho) = p\rho \oplus (1-p)\operatorname{Tr}\rho\,|\mathrm{er}\rangle\langle\mathrm{er}|. \tag{27}$$

The erasure flag is orthogonal to the transmitted input. The quantum-capacity formula is $Q(\mathcal{E}_{d,p}) = \max\{0, 2p-1\}\log_2 d$ [16]. In particular the critical value $p = 1/2$ has zero quantum capacity.

Proposition 4.1 (Half-erasure positivity principle)**.** *Suppose that for some finite b there are an isometric encoder $V : \mathbb{C}^2 \to A^{\otimes b}$ and a receiver channel $\mathcal{D}$ with*

$$\mathcal{D} \circ L^{\otimes b} \circ \mathrm{Ad}_V = \mathcal{E}_{2,1/2}, \qquad \mathrm{Ad}_V(X) = VXV^*. \tag{28}$$

Then $P(L) > 0$ if and only if $Q(L) > 0$.

The proposition separates two ingredients: the established activation identity, and the realization of its helper from further uses of L. The latter is necessary for a statement about unassisted capacity.

Lemma 4.2 (Finite Smith–Yard identity)**.** *Let a finite ensemble $\{p_x, \rho_x\}$ for a channel K have private information $\eta = \chi(X:B) - \chi(X:E) > 0$. Put $d = \sum_x \operatorname{rank}\rho_x$. There is a fixed joint input state ω, independent of p, such that for every $0 \leq p \leq 1$,*

$$\begin{aligned} I_{\mathrm{c}}(K \otimes \mathcal{E}_{d,p}, \omega) &= p[H(X) - \chi(X:E)] + (1-p)[\chi(X:B) - H(X)] \\ &= \frac{\eta}{2} + (p - \tfrac{1}{2})[2H(X) - \chi(X:B) - \chi(X:E)]. \end{aligned} \tag{29}$$

In particular it is positive for p sufficiently close to $1/2$.

Proof. Purify each ρ_x to $|\psi_x\rangle_{AF_x}$, using orthogonal auxiliary spaces of dimension rank ρ_x. With $F = \bigoplus_x F_x$, prepare

$$|\Psi\rangle_{R_0AF} = \sum_x \sqrt{p_x}|x\rangle_{R_0}|\psi_x\rangle_{AF_x}.$$

The channel input is $\omega = \mathrm{Tr}_{R_0}\Psi$. When F is transmitted, $S(BF) = H(X) + \sum_x p_x S(E_x)$ and Eve has entropy $S(\bar{E})$. When F is erased, Bob has entropy $S(\bar{B})$ and $S(EF) = H(X) + \sum_x p_x S(B_x)$. The input-independent branch entropy appears at both outputs and cancels. Averaging these two cases gives (29). Its final bracket lies in $[0, 2H(X)]$. At $p = 1/2$ the identity is exactly the finite Smith–Yard calculation [10, Supporting material, Eqs. (S1)–(S4)]. □

Lemma 4.3 (A fixed helper from qubit half erasures)**.** *For every fixed d, there are isometries $W_m : \mathbb{C}^d \to (\mathbb{C}^2)^{\otimes m}$ and receiver channels $\mathcal{D}_m$ such that*

$$\mathcal{D}_m \circ \mathcal{E}_{2,1/2}^{\otimes m} \circ \mathrm{Ad}_{W_m} = \mathcal{E}_{d,p'_m}, \qquad p'_m \longrightarrow \tfrac{1}{2}. \tag{30}$$

The codes and decoders can be found by finite search. Their logical rate $\log_2 d/m$ tends to zero.

Proof. Embed the fixed input in $k' = \lceil \log_2 d \rceil$ logical qubits. For an $[[m, k']]$ stabilizer code, consider masks with at least

$$t_m = \lceil (m + k')/2 \rceil + \lceil m^{1/4} \rceil$$

survivors. By Lemma 3.2, the average probability mass of nongood masks in this set, over the random code, is at most $\epsilon_m = 2^{1-2\lceil m^{1/4} \rceil}$. Some fixed code consequently has good-mask mass p'_m satisfying

$$p_m - \epsilon_m \le p'_m \le p_m, \qquad p_m = \Pr[\mathrm{Bin}(m, 1/2) \ge t_m].$$

On those good masks the receiver recovers the full logical input exactly. On every other mask it outputs the erasure flag. These are trace-preserving branch decoders; the flags are actual outputs, and every mask has input-independent probability. Forgetting the mask record therefore gives exactly (30). The decoder from the embedded logical space back to $\mathbb{C}^d$ is completed trace-preservingly outside the embedding.

The same central-binomial estimate used above gives $p_m \to 1/2$ and thus $p'_m \to 1/2$. For each stabilizer and mask the normalizer certificate is a finite linear-algebra test. Enumerating stabilizer codes and masks finds one with the stated rational probability bound; no efficiency assertion is needed. □

Proof of Proposition 4.1. Suppose $P(L) > 0$. The regularized formula supplies a finite block $K = L^{\otimes n}$ and finite ensemble with $\eta > 0$. Fix the activation input of Lemma 4.2, hence also its finite helper dimension d. Lemma 4.3 and (28) realize $\mathcal{E}_{d,p'_m}$ using bm further slots of L and receiver processing. For sufficiently large finite m, equation (29) is strictly positive.

The sender embeddings are isometric. Thus the purification reference of this effective input is also a purification reference for the raw physical input to $L^{\otimes(n+bm)}$. Coherent information equals $-S(R_0|B)$ and cannot increase under receiver processing. The raw block therefore has positive coherent information as well, proving $Q(L) > 0$ by (6). The converse is $Q \le P$. □

For $\mathcal{C}_z$, fix A pure and discard its branch payloads at Bob. The auxiliary input Z is then transmitted perfectly in branch one and erased in branch zero: this is exactly $\mathcal{E}_{z,1/2}$. Restricting to two coordinates gives (28) with $b = 1$. Proposition 4.1, together with Section 3 and Proposition 2.3, completes the capacity equivalence in Theorem 1.1.

The order of choices matters: first fix the private witness and its finite helper, then choose enough half-erasure slots to make (29) positive. All helper slots count in the resulting finite blocklength. The construction stays below transmission probability 1/2 and uses vanishing

logical helper rate, consistently with the no-cloning obstructions of [17, 18]. Because the decoded helper is an exact erasure channel, its effect on coherent information is the explicit affine expression (29); no separate approximation of its environment is required.

5 From channels to one-way distillation

For a bipartite state ω_{AB}, $D_{\rightarrow}(\omega)$ is the asymptotic rate of maximally entangled bits distillable with public messages only from Alice to Bob. Likewise $K_{\rightarrow}(\omega)$ is the ordinary distillable secret-key rate with the same communication direction. Eve holds a purification and every public message. Alice and Bob's local ancillary systems and discarded laboratory registers remain trusted; no initial entanglement or key is supplied. The two-way versions are denoted $D_{\leftrightarrow}, K_{\leftrightarrow}$.

A Choi state records the channel acting on half of a maximally entangled pair. Its input marginal is maximally mixed, while a positive capacity witness may require a very different input. Steering bridges the mismatch: Alice filters her half so that the other half has the desired input marginal. The filter can be unlikely, but its success probability is strictly positive for every fixed finite witness. She can announce success forward, so ordinary one-way distillation retains a positive rate after failures are charged.

The converse uses the same identity in reverse. Each public branch of a one-way protocol on Choi states specifies an ordinary channel input or input ensemble. Its information cannot exceed the best block channel information. Thus steering preserves positivity in both directions without asserting equality of the rates.

5.1 The standard one-way coding formulas

For a bipartite state write $I_{\mathrm{c}}(A\rangle B)_{\omega} = S(B)_{\omega} - S(AB)_{\omega}$. The one-way coding theorems of Devetak and Winter [3, Theorems 8 and 13] give

$$D_{\rightarrow}(\omega) = \sup_{n\geq 1} \frac{1}{n} \sup_{\{\mathcal{M}_t\}} \sum_t p_t I_{\mathrm{c}}(A'\rangle B^n)_{\omega_t}, \tag{31}$$

$$K_{\rightarrow}(\omega) = \sup_{n\geq 1} \frac{1}{n} \sup_{\{Q_{x,t}\}} [I(X:B^n|T) - I(X:E^n|T)]. \tag{32}$$

In the first line $\{\mathcal{M}_t\}$ is a finite Alice instrument on A^n, with normalized branch states ω_t and probabilities p_t. Bob receives the announced t. In the second line $\{Q_{x,t}\}$ is a finite POVM on A^n; Alice keeps x and announces t. Any classical preprocessing can be folded into these effects. The state used to evaluate Eve's information is obtained by measuring Alice's part of a purification of $\omega^{\otimes n}$.

The optimization includes arbitrary local preprocessing. The coding theorems supply the asymptotic statements; convexity and steering give the finite witnesses needed below. For completeness, the usual finite-key converse has a dimension bound involving only the key: if an M-valued key has correctness error ϵ, and let δ be half the trace norm distance from a uniform key independent of Eve and the public transcript. Then

$$I(X:B^n|T) - I(X:E^n|T) \geq (1-2\delta-\epsilon)\log_2 M - (1+\delta)h_2\left(\frac{\delta}{1+\delta}\right) - h_2(\epsilon).$$

Here $h_2(u) = -u\log_2 u - (1-u)\log_2(1-u)$ is binary entropy, with the continuous endpoint values. The bound follows from Fano's inequality and conditional-entropy continuity [22, Lemma 2]; the continuity dimension is the key alphabet, independently of the size of the public transcript.

Lemma 5.1 (Finite one-way witnesses)**.** *For every finite bipartite state ω:*

1. *$D_{\to}(\omega) > 0$ if and only if for some finite n and one nonzero Alice filter F the normalized state*
$$\tau = \frac{(F \otimes I)\omega^{\otimes n}(F^* \otimes I)}{p}, \qquad p = \mathrm{Tr}[(F \otimes I)\omega^{\otimes n}(F^* \otimes I)] > 0$$
has $I_{\mathrm{c}}(A'\rangle B^n)_\tau > 0$. The filter may be chosen square on A^n and rescaled to $F^*F \le I$.
2. *$K_{\to}(\omega) > 0$ if and only if some finite n and finite POVM $\{Q_{x,t}\}$ give $I(X : B^n|T) - I(X : E^n|T) > 0$.*

Proof. The single-Kraus refinement in [3, Theorem 13] preserves or increases the instrument average in (31). A positive average therefore has a positive single-filter branch. Its supported output dimension is at most $\dim A^n$, allowing a square representative up to an output isometry.

Conversely, apply a physical filter independently to n-copy blocks and announce success forward. Retain a deterministic number of successes slightly below the typical fraction p. Insufficient successes have probability tending to zero, and hashing on the retained states yields
$$D_{\to}(\omega) \ge \frac{p}{n} I_{\mathrm{c}}(A'\rangle B^n)_\tau > 0. \tag{33}$$
Thus this is an ordinary rate, with failures charged. The key assertion is exactly the finite-positive-witness content of (32); its sufficiency is conditional one-way key coding on independently repeated blocks. □

5.2 The Choi positivity bridges

Let $L : A_0 \to B$ have input dimension d, Stinespring isometry $V : A_0 \to BE$, and normalized Choi state
$$J_L = (\mathrm{id}_A \otimes L)(\Phi_d), \qquad |\Phi_d\rangle = d^{-1/2} \sum_{i=1}^{d} |i\rangle_A |i\rangle_{A_0}. \tag{34}$$
The purification is $(I \otimes V)|\Phi_d\rangle$. The transpose below always uses this fixed basis. For a positive effect Q, the unnormalized conditional receiver–environment state is
$$\mathrm{Tr}_A[(Q \otimes I)(I \otimes V)\Phi_d(I \otimes V^*)] = d^{-1} V Q^T V^*. \tag{35}$$
The same identity applies to blocks, with d replaced by $D = d^n$. This is ensemble steering [4] in the maximally entangled basis. It gives Bob's and Eve's states simultaneously, including for mixed input letters. The failure effect in the proof below completes any desired scaled ensemble to the maximally mixed input marginal.

Theorem 5.2 (Choi positivity and rate bounds)**.** *For every finite memoryless channel L,*
$$D_{\to}(J_L) \le Q(L), \qquad D_{\to}(J_L) > 0 \iff Q(L) > 0, \tag{36}$$
$$K_{\to}(J_L) \le P(L), \qquad K_{\to}(J_L) > 0 \iff P(L) > 0. \tag{37}$$
More precisely, if a block input ρ has positive coherent information, then
$$D_{\to}(J_L) \ge \frac{I_{\mathrm{c}}(L^{\otimes n}, \rho)}{n d^n \|\rho\|_\infty}. \tag{38}$$
If a finite ensemble with average ρ has positive private information Δ for that block, then
$$K_{\to}(J_L) \ge \frac{\Delta}{n d^n \|\rho\|_\infty}. \tag{39}$$

Proof. Set $D = d^n$ and $\lambda = \|\rho\|_\infty$. Alice's filter

$$F = \sqrt{\rho^T}/\sqrt{\lambda}$$

is a contraction. Its probability on the Choi reference is $s = 1/(D\lambda)$, and its successful prechannel vector is

$$\sqrt{D\lambda}(F \otimes I)|\Phi_D\rangle = (I \otimes \sqrt{\rho}) \sum_i |i\rangle|i\rangle. \tag{40}$$

This vector purifies ρ. Since the filter commutes with the channel on the other side, its normalized output has coherent information $I_c(L^{\otimes n}, \rho)$. Equation (33) proves (38). Every positive quantum capacity has such a finite block witness.

For a private ensemble $\{p_x, \rho_x\}$ with average ρ, use the POVM

$$Q_x = p_x \rho_x^T/\lambda, \qquad Q_f = I - \rho^T/\lambda. \tag{41}$$

It is positive and sums to I. Formula (35) gives the successful BE ensemble

$$\frac{p_x}{D\lambda}\, V^{\otimes n} \rho_x (V^{\otimes n})^*.$$

Its total probability is again $s = 1/(D\lambda)$. Alice announces only success or failure, using a constant private label on failure. The conditional information difference is exactly $s\Delta$. Equation (32) proves (39). In particular $P(L) > 0$ implies $K_\to(J_L) > 0$. We have not refined mixed input letters into publicly revealed pure components.

For the upper entanglement bound, consider any instrument on $J_L^{\otimes n}$. Its prechannel branch states are

$$\theta_t = (\mathcal{M}_t \otimes \mathrm{id})(\Phi_D)/p_t.$$

Each branch output is $(\mathrm{id} \otimes L^{\otimes n})(\theta_t)$. Decomposing θ_t into pure joint reference–input states and using convexity shows that its coherent information is at most $Q^{(1)}(L^{\otimes n}) = \max_\rho I_c(L^{\otimes n}, \rho)$. Each pure component purifies its channel-input marginal, which may be mixed. Hence every instrument average in (31) is bounded by this same quantity. Regularization proves $D_\to(J_L) \le Q(L)$.

For the key upper bound, let $\{Q_{x,t}\}$ be any POVM on the block Choi reference. For every outcome of positive probability put

$$p_{x,t} = \operatorname{Tr} Q_{x,t}/D, \qquad \rho_{x,t} = Q_{x,t}^T/\operatorname{Tr} Q_{x,t}. \tag{42}$$

By (35), each conditional ensemble $\{p(x|t), \rho_{x,t}\}_x$ is a valid ordinary block input ensemble, with exactly the same Bob and Eve states. If its private-information gap is Δ_t, then

$$I(X : B^n|T) - I(X : E^n|T) = \sum_t p_t \Delta_t \le P^{(1)}(L^{\otimes n}). \tag{43}$$

Regularization gives $K_\to(J_L) \le P(L)$. If the average is positive, one can fix a positive branch t in the code design and use its ensemble directly; no transcript must be transmitted through a free side channel. This is the same transcript-averaging principle used in the standard private-transmission/key-generation converse [2, Section 2]. The two upper bounds finish the positivity equivalences. □

The probability $1/(d^n\|\rho\|_\infty)$ expresses the price of changing a maximally mixed input into the desired one. It is one when $\rho = I/d^n$, and d^{-n} for a pure input. This gives a concrete reason for the distinction between positivity and equality of rates. For key distillation, mixed letters remain mixed: Alice's trusted measurement residual stays in her laboratory, as required by the resource convention stated above.

5.3 Uniform two-way entanglement in every compiled state

The universal compiler gives a particularly strong separation of one-way and two-way tasks. Its normalized Choi input is the product $\Phi_{\dim A} \otimes \Phi_z$. More explicitly its two Bob branches are

$$J_{\mathcal{C}_z} = \tfrac{1}{2} J_M \otimes \frac{I_{Z'}}{z} \;\oplus\; \tfrac{1}{2} J_{N^c} \otimes \Phi_{Z'Z},$$

where the direct sum is on Bob and Z' is Alice's auxiliary reference. Alice traces out the reference of A. Bob discards the M or N^c payload and keeps the transmitted Z or an erasure flag, according to his branch. These local channels produce exactly

$$J_{\mathcal{E}_{z,1/2}} = \tfrac{1}{2}\Phi_z + \tfrac{1}{2}\frac{I_z}{z} \otimes |\mathrm{er}\rangle\langle \mathrm{er}|. \tag{44}$$

Bob can announce the successful positions backward. Each surviving pair is maximally entangled of dimension z, and their typical fraction is $1/2$. Consequently

$$K_{\leftrightarrow}(J_{\mathcal{C}_z}) \geq D_{\leftrightarrow}(J_{\mathcal{C}_z}) \geq \tfrac{1}{2}\log_2 z. \tag{45}$$

For $z = 4$ the bound is already one ebit per copy, without adding a separate erasure factor. The full Alice marginal is exactly $I/(z \dim A)$. Taking z to be a sufficiently large fixed power of two gives any prescribed fixed lower bound in (7). The backward announcement is used solely for this two-way bound, never in the one-way bridges or unassisted channel codes.

6 Effective descriptions and exact-zero classification

The channel itself is computable without finding a violating reference, sharing code, or positive-capacity block. Those objects enter only the proof of its behavior. This separation is what allows the compiler to transfer an undecidable comparison.

6.1 Rational complements and the compiler map

A rational-complex description means that every matrix entry has rational real and imaginary parts, given exactly. Validity of a finite Choi description is decidable: Hermiticity, positivity, and the partial-trace identity are exact rational conditions. We work throughout on this valid-description domain.

Lemma 6.1 (Exact rational complements)**.** *Every positive semidefinite rational-complex matrix has an effectively computable rational-complex Gram factor if redundant columns are allowed. Hence every rational-complex channel has a computable rational-complex Kraus family and a rational-complex complement.*

Proof. Rational Hermitian LDL^* elimination, with permutation pivots when needed, writes the matrix as $\sum_j q_j v_j v_j^*$, where $q_j > 0$ is rational and v_j is rational complex. A zero diagonal in a positive semidefinite remainder has zero row and column and is omitted. Write $q_j = a_j/b_j$ with positive integers a_j, b_j and enumerate a four-square representation $a_j b_j = \sum_{t=1}^{4} z_{jt}^2$. Lagrange's theorem guarantees termination, and

$$q_j v_j v_j^* = \sum_{t=1}^{4} [(z_{jt}/b_j)v_j][(z_{jt}/b_j)v_j]^*.$$

These are rational Gram columns.

Apply this construction to the unnormalized Choi matrix. Unvectorizing its Gram columns gives rational Kraus matrices K_i, with $\sum_i K_i^* K_i = I$, by the Choi correspondence [28]. The isometry $V = \sum_i K_i \otimes |i\rangle$ has rational entries and its complement has rational coefficients

$$(L^c(X))_{ij} = \mathrm{Tr}(K_i X K_j^*).$$

All steps are finite and effective; minimal Kraus rank is unnecessary. □

Let $a = \dim A$, let b_M, b_N be the original output dimensions, and let e_M, e_N be the chosen Kraus-index dimensions. The compiler has

$$\dim A_{\mathcal{C}_z} = za, \qquad \dim B_{\mathcal{C}_z} = b_M + ze_N, \qquad \dim E_{\mathcal{C}_z} = ze_M + b_N. \tag{46}$$

Its two direct-sum linear maps have rational coefficients by Lemma 6.1. The $1/\sqrt{2}$ amplitudes in a balanced Stinespring dilation disappear upon either partial trace; they do not introduce irrational Choi entries. This proves the effective-description assertion of Theorem 1.1.

6.2 Finite positive witnesses are enumerable

We give the semidecision arguments explicitly, since undecidability alone does not identify the precise arithmetical classification. A semidecision for positivity halts on every positive instance and never accepts a zero one; it need not halt on zero instances.

For a channel, enumerate all blocklengths and rational input states, for example the dense family $GG^*/\mathrm{Tr}(GG^*)$ for nonzero rational-complex matrices G. Using the rational complement above, compute certified intervals for the resulting coherent information, and accept when a lower bound is strictly positive. Finite-dimensional entropy is computable also at zero eigenvalues: $-x\log_2 x$ has an effective continuous extension there. Dovetailing all candidates and precision levels gives a sound search. Density, continuity, and (6) make it complete for $Q > 0$. For $P > 0$, also enumerate finite ensemble sizes and rational probabilities, and use (5). No bound on the witnessing blocklength is needed.

For a rational bipartite state, enumerate blocklengths and rational square Alice filters. Compute their unnormalized output and its probability exactly; skip zero-probability candidates. A nonzero filter can be rescaled by a sufficiently small positive rational to be a contraction without changing its normalized output. Search for a certified positive coherent information. The first part of Lemma 5.1 makes this sound and complete: a positive real filter can be rationally approximated while keeping both its probability and strict information gap positive.

For key, enumerate blocklengths, finite classical alphabets, and rational-complex POVMs $\{Q_{x,t}\}$. Positivity and exact sum to I are decidable. Compute a purification of the input state by vectorizing its positive square root; the entries need only be computable, not rational. Search certified entropy intervals for a positive conditional information difference in (32).

Rational POVMs suffice for completeness. Given a positive witness with m outcomes, first replace its effects by $Q_j^{(\eta)} = (1-\eta)Q_j + \eta I/m$ for small $\eta > 0$. Every effect becomes strictly positive, while continuity preserves the gap. Approximate the first $m-1$ effects by rational Hermitian matrices and set the last to I minus their sum. A sufficiently accurate approximation preserves positivity of all effects and the gap. Thus each of the four positivity sets in Theorem 1.2 belongs to RE independently of the hardness reduction.

6.3 The reductions and the two-way promise

Apply (12) to a Turing machine T and compute $L_T = \mathcal{C}_z(M_T, N_T)$, with any fixed integer $z \geq 4$. The universal equivalence gives the simultaneous dichotomy

$$\begin{aligned} &T \text{ does not halt}: && Q(L_T) = P(L_T) = 0, \\ &T \text{ halts}: && Q(L_T) > 0 \text{ and } P(L_T) > 0. \end{aligned} \tag{47}$$

The map is computable on exact rational descriptions by Lemma 6.1. It proves RE-hardness of both channel positivity sets. Their membership just established proves RE-completeness; complementing within the valid-description domain proves coRE-completeness of the exact-zero sets.

The normalized Choi map is also rational and effective:

$$J_L = \frac{1}{\dim A_L} \sum_{i,j} |i\rangle\langle j| \otimes L(|i\rangle\langle j|).$$

Set $\omega_T = J_{L_T}$. Theorem 5.2 converts (47) into

$$T \text{ halts} \quad \Longleftrightarrow \quad D_{\rightarrow}(\omega_T) > 0 \quad \Longleftrightarrow \quad K_{\rightarrow}(\omega_T) > 0.$$

Thus the same membership and complement arguments prove the unrestricted state classifications.

Finally, for a prescribed rational $r > 0$, fix once and for all

$$k_r = \max\{2, \lceil 2r \rceil\}, \qquad z = 2^{k_r}.$$

Every target has exactly maximally mixed Alice marginal and, by (45), $K_{\leftrightarrow}(\omega_T) \geq D_{\leftrightarrow}(\omega_T) \geq k_r/2 \geq r$. This is hardness under the joint promise (7); the semidecision procedures still work on all promised inputs. No decision procedure for membership in the two-way promise domain is asserted. This completes Theorem 1.2.

The resulting distinction is exact zero versus strict positivity in variable dimensions. Reference transmission, activation and steering each preserve a finite positive witness, with costs depending on that witness. Those costs explain why the compiler yields no uniform positive capacity or one-way distillation gap, and why the entropy-defect inapproximability theorem of the companion does not transfer to these rates. Meanwhile every hard Choi state retains the explicit two-way resource (45). The undecidability input throughout is precisely the complete-order reduction of Theorem 2.2.

A Why complementing reverses complete order

For completeness we explain the purification argument behind (11). Its one input from the companion paper [1, Theorem 3.1] is the following precisely stated equivalence: complete relative-entropy order is equivalent to complete SLD order. The latter compares

$$g_\rho(X) = \mathrm{Tr}[X\Omega_\rho^{-1}(X)], \qquad \Omega_\rho(H) = \tfrac{1}{2}(\rho H + H\rho), \tag{48}$$

after every finite reference extension, at every faithful state ρ and traceless Hermitian tangent X. In an eigenbasis of a possibly singular baseline,

$$g_\rho(X) = \sum_{i,j:\rho_i+\rho_j>0} \frac{2|X_{ij}|^2}{\rho_i + \rho_j}, \tag{49}$$

provided the kernel-to-kernel block of X vanishes; otherwise the value is $+\infty$. Cross terms between support and kernel are retained. All inverses use the effective output support $\mathrm{supp}\,\Phi(I)$.

The key identity uses a dual input that is fixed before the channel is chosen. Let $\rho_{RA} > 0$, $\mathrm{Tr}\,X = 0$, set $H = \Omega_\rho^{-1}(X)$, and choose a purification $|\psi\rangle_{RAF}$. Define

$$\Gamma = \mathrm{Tr}_R\,\psi, \qquad \xi = \frac{i}{2}\,\mathrm{Tr}_R[H \otimes I_F, \psi].$$

For every channel $\Phi : A \to B$ and genuine complement,

$$g_{\Phi_R(\rho)}(\Phi_R(X)) = g_\rho(X) - g_{\Phi_F^c(\Gamma)}(\Phi_F^c(\xi)). \tag{50}$$

The same (Γ, ξ) works for every Φ. It expresses the information lost at the receiver as a complementary phase-direction quantity, rather than comparing unrelated state-dependent purifiers. This is the purification geometry associated with quantum information–disturbance identities; related metrological formulations appear in [29].

Here are the details, including singular environmental states. For a Stinespring isometry put

$$|\Psi\rangle = (I_R \otimes V \otimes I_F)|\psi\rangle, \qquad |\dot\Psi\rangle = (I_R \otimes V \otimes I_F)(H \otimes I_F)|\psi\rangle/2.$$

Then $4\|\dot\Psi\|^2 = g_\rho(X)$ and $\langle\Psi|\dot\Psi\rangle = 0$. For the partition $RB : EF$, set

$$A_0 = \operatorname{Tr}_{RB} \Psi = \Phi_F^c(\Gamma), \qquad B_0 = i \operatorname{Tr}_{RB}(|\dot\Psi\rangle\langle\Psi| - |\Psi\rangle\langle\dot\Psi|) = \Phi_F^c(\xi).$$

Changing the purification by an environmental unitary leaves the receiver tangent unchanged. Its least squared speed is

$$g_{\Phi_R(\rho)}(\Phi_R(X)) = \min_{J=J^*} 4\|\dot\Psi - i(I \otimes J)\Psi\|^2. \tag{51}$$

To verify this directly, write a purification as a coefficient matrix P, with $PP^* = \sigma$. A derivative Q has output tangent Y exactly when $QP^* + PQ^* = Y$. The matrix $Q_0 = \Omega_\sigma^{-1}(Y)P/2$ satisfies this condition and is real-orthogonal to every K with $KP^* + PK^* = 0$. It therefore minimizes the norm, with $4\|Q_0\|^2 = g_\sigma(Y)$. In Schmidt coordinates $P = [\sqrt{\sigma}\ \ 0]$, every such K is iPJ for some Hermitian J, including its unrestricted second block. This proves (51).

Expanding that minimum gives $g_\rho(X) + 4\operatorname{Tr}(A_0 J^2) + 4\operatorname{Tr}(B_0 J)$. The normal equation is $\Omega_{A_0}(J) = -B_0/2$. Completing the square proves (50). The phase path

$$\operatorname{Tr}_R(e^{iuH/2}\psi e^{-iuH/2})$$

is positive for both signs of u; its derivative and every channel output derivative have zero kernel-to-kernel block. Thus the supported form (49) applies and is finite.

An SLD comparison at faithful inputs extends to these dual pairs. Replace them by $((1-\epsilon)\Gamma + \epsilon I_{AF}/\dim(AF), (1-\epsilon)\xi)$ and let $\epsilon \downarrow 0$. The supported spectral formula converges: denominators involving the support stay positive, and the vanishing kernel-to-kernel tangent block removes the possible divergence. Equivalently, in the support/kernel decomposition the new kernel eigenvalues are of order ϵ, while the corresponding tangent block is of order ϵ after diagonalization, so its contribution vanishes.

Now assume $\Phi \succeq^{\mathrm{c}}_{\mathrm{SLD}} \Psi$. Apply the common identity to Φ^c, Ψ^c at the same arbitrary faithful primal input. Their double complements are Φ, Ψ up to output isometries, and the hypothesis compares their common dual pair. Subtracting from $g_\rho(X)$ gives $\Psi^c \succeq^{\mathrm{c}}_{\mathrm{SLD}} \Phi^c$. Applying this implication again to the complements gives the converse. The companion's SLD/relative-entropy equivalence then proves $\Phi \succeq^{\mathrm{c}}_D \Psi \iff \Psi^c \succeq^{\mathrm{c}}_D \Phi^c$. The full metric equivalence, unlike this short purification argument, is proved in the companion and is not an assertion imported from a theorem that establishes only one direction.

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